



REQUEST FOR PROPOSAL

MEMBER APPLICATION SOLUTION

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1. INTRODUCTION

This document is a Request for Proposal (“RFP”) for the design and implementation of a Member Application Solution. The member application solution must meet the functionality, user and technical requirements. The contract term will cover the design, development, testing, and acceptance of the revised member application and architecture solution.

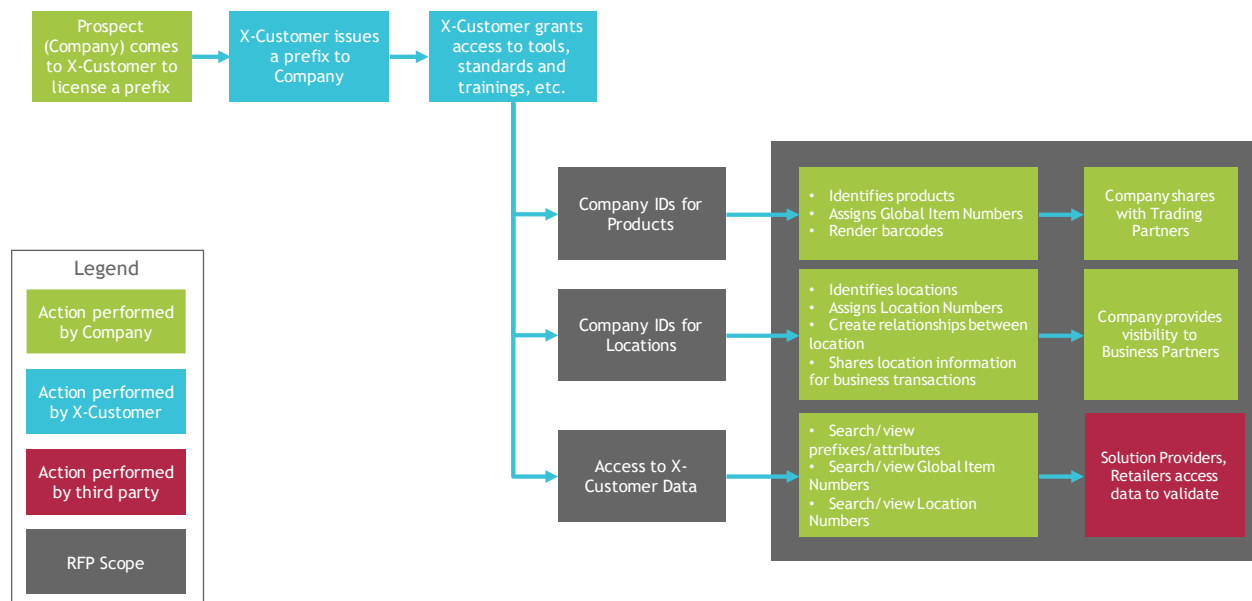
X-Customer serves more than 200,000 businesses in 25 industries in the United States by facilitating industry initiatives, administrating the X-Customer system of standards, providing education and support, and connecting communities through events and online forums.

1.1. STATEMENT OF WORK BUSINESS OVERVIEW

As part of its mission to “generate supply chain visibility, efficiency, safety, and collaboration” X-Customer sets the standards for and enables its members to identify products and/or locations with numeric indicators that can be used to facilitate the supply chain process.

For example, a X-Customer member can create a 12- or 14-digit number to identify a product and translate the number to a barcode format that can be used in retail transactions. The project scope includes the design and development of a B2B application solution used to create, manage, and share data related to these numeric indicators. The application solution will be used by businesses that range from one -person start-up companies to large corporations to facilitate their supply chain activities.

Figure 1 shows the process for creating and sharing these numbers.



The process starts with new membership; X-Customer licenses a 6-9 digit prefix, which is the baseline for a company to enumerate their products and/or locations. Onboarding members and managing prefix attributes are outside of the scope of this project.

Once a company has licensed a prefix, they can create additional numeric identifiers for their products or locations and add identifying attributes to each.

- Products are identified by 12- or 14-digit Global Item Numbers (GIN) that can be represented visually by a barcode.
- Locations are identified by 13-digit Location Number (LN). Location Numbers are similar in format to Entity Global Item Numbers but have a different set of attributes.

GIN and LNs are comprised of the prefix + a series of unique numbers. For example, 9-digit prefix can be used to create 100 unique 12-digit GIN by adding unique numbers for the last three digits. The same prefix can be used to create 1000 13-digit LNs by adding unique numbers for the last four digits.

GIN example:

00012345678905

Barcode example:



1.2. PROJECT GOALS

Currently, X-Customer has several applications available to their members. To support the needs of their growing member base and increase the compliance of global standards, X-Customer intends to implement web-based application solution that is built on a scalable, sustainable architecture.

This project will design and build an integrated solution with the following modules:

- Create and manage GIN for Products
- Create and manage LN for Locations
- Access data related to Prefixes, GIN, and LN created and shared by others

Each module in the integrated solution will use the same data and system architectures and run on the same development platform. Final front end design will define whether these modules will appear as a single application or as a suite of related applications.

The project goals are:

1. **Design and implement a web-based application solution** that extends application functionality and enhances the end user experience through intuitive navigation, work process flows, and new look and feel.
2. **Design and implement an application architecture** that is able to support the application requirements and align with the existing enterprise architecture. The application architecture should be designed and implemented in a way that is maintainable by X-Customer resources and is scalable and sustainable given the projected member and application growth.
3. **Create efficiencies and cost reductions** for future enhancements through a standard application framework and scalable application architecture.

Future expansion may include creation, management, and sharing of data other than GIN or LN as well as increasing data sharing privileges to the field level. While these expansions are outside the scope of the project, the final design should be scalable to accommodate that.

1.3. PROJECT STRUCTURE

To complete the project goals, the Member Application Solution project is divided into four primary components:

1. Requirements and Design
2. Development and Testing
3. Implementation
4. Post-production Support and Transition

X-Customer prefers to select a vendor who will perform all four functions for both the web -based application solution and the application architecture. If, however, a respondent chooses to partner with a subcontractor(s) to complete any part of the project, all components, including pricing for each component, should be structured similarly and submitted as one response.

2. REQUIREMENTS

2.1. GENERAL REQUIREMENTS

The application solution will include functionality available to all members based on permissions, regardless of which module(s) they use. Many of the functions relate to data and functionality across modules. All functionality must be validated against X-Customer Standards and business rules, which will be defined as part of the project. High level requirements dictate that the system will:

ID	Requirement Description
G-1.	Allow assignment of user roles and tasks
G-1.1.	Each member is assigned a company administrator with the ability to add/edit other users in their company
G-1.2.	Company admin can assign roles and tasks to users
G-1.3.	User roles include Super-User, Approver, Editor, View-only; other roles may be added
G-1.4.	Support customer-level “Super-User” role that has access to manage all data
G-2.	Present a role-driven UI based on user role
G-3.	Implement a workflow for new and edited records
G-3.1.	Company admin can designate Editor and Approver roles and rules for record approval
G-3.2.	Editors and Approvers must be notified of a change in approval status
G-3.3.	Users must be able to easily find records requiring their action
G-4.	Allow a user’s password to be re-set by the user, the company admin, or the X-Customer Help desk
G-5.	Lock records so that only one user at a time can edit a record
G-6.	Display a user “dashboard” or start screen with notifications, reports, prefix data, and other information
G-6.1.	Prefix capacity counter should display how many numeric indicators (GINs or LNs) have been used and how many are available
G-7.	Allow users to generate reports
G-7.1.	Customize, schedule, run, and view reports for adds, changes, and deletes
G-7.2.	View usage reports by hour, week, day, month, year; current reporting utilizes 3Scale
G-7.3.	Run report data history
G-7.4.	Run audit reports
G-8.	Provide notifications to users in various formats (e.g., email, SMS, and/or onscreen)
G-8.1.	Global notifications (e.g. User will be notified of scheduled server down time)
G-8.2.	Member-specific notifications (e.g. User can opt to be notified when records are updated or when new records are added)
G-8.3.	Member-to-member/ X-Customer notifications (e.g. User can challenge record data)
G-8.4.	Other notifications may be added

G-8.5.	User can set communication preferences for the method and frequency they receive notifications outside the system
G-9.	Include ability for users to send application feedback to X-Customer, routed to a single X-Customer contact or feedback tracking system
G-10.	Display contextual help to the user
G-10.1.	Help content may be customized by industry
G-10.2.	Help content must be maintained by X-Customer nontechnical resources
G-11.	Link to training, such as external videos, webinars, quick-start guides within the application
G-11.1.	Development of training materials is outside the scope of this project
G-12.	Allow import and export of records, as referenced in Product, Location, and Data Access modules
G-12.1.	Import and Export formats include at least Excel, CSV, Numbers, XML, tab- delimited, iDocs, and through the API
G-12.2.	Must include PC and Mac formats
G-12.3.	User can select one or more records to be imported or exported
G-12.4.	All records must be validated upon import
G-13.	Include a publish (select for sharing by data owner) and Subscribe (request viewing by data consumers) model with multiple share/view functions:
G-13.1.	The data owner must be able to publish records at any level in the hierarchy
G-13.2.	The data owner must be able to publish (grant access to view) data for a basic or full set of record attributes
G-13.3.	The data owner must be able to publish data for a group of records
G-13.4.	The data owner must be able to share records with an individual, a user-defined group, a controlled group managed by X-Customer, or the public
G-13.5.	Based on subscription, a data consumer must be able to view published data
G-13.6.	Based on subscription, a data consumer must be able to request access to view data from data owner
G-13.7.	The data owner must have the ability to approve/reject view requests

2.2. PRODUCT REQUIREMENTS

The Product Module is used by data owners to create and manage GIN data related to their own products and to permission data to be shared in the Access Data module. Users range from entrepreneurs launching their first product to large companies with a high volume of inventory. Many users will use the tool only when they launch a new product. High level requirements enable the user to:

ID	Requirement Description
P-1.	Create GIN records individually, via import, through a step-by-step wizard, or by cloning an existing record. Records will include required and non-required attributes that may vary by industry
P-1.1.	Incomplete records can be saved as Draft
P-2.	Assign a GIN to a record automatically or manually at any point during record creation; a check-digit will be automatically assigned to the GIN
P-2.1.	Users can opt for auto-assign or manual
P-2.2.	User can reserve GINs to hold out from auto-assign
P-3.	Set a preference for how GINs will be assigned (e.g., auto-assign or manual)
P-4.	Edit record attributes before finalizing
P-5.	Edit record attributes manually or via import
P-5.1.	Some attributes cannot be edited, based on status (e.g., if a product has gone to market)
P-5.2.	If a GIN changes but the product attributes did not, a reference should be made back to the old number
P-6.	Upload product images and add them to records
P-7.	View, filter, and sort all records in a single view
P-7.1.	Select individual records to take any action (e.g., edit, export, etc.)
P-7.2.	Select multiple records to take any action and apply the same action to all selected records
P-8.	View, filter, and sort all GINs in a single view
P-9.	View potential duplicate records and easily remove or edit duplicate records
P-9.1.	System will alert user of possible duplicate records during creation process as well as run duplicate reports for completed records
P-10.	View and manage the status of a record or GIN
P-10.1.	The system or user will assign a status to each GIN (current statuses are Reserved, In Use, For Reuse, Available); GIN status may update to reflect record status
P-10.2.	The system or user can assign a status to each record
P-10.3.	If a GIN is no longer in use, display the date taken out of use and when it will be available for re-use; rules may vary by industry and product type
P-11.	Create, edit, and view a hierarchy of GINs (e.g., assigning an item type “each” to an item type “case” and a “case” to a “pallet”) manually, via import, or via a visual format such as drag-and-drop
P-11.1.	Hierarchies can have pre-defined levels

P-11.2.	Imports can include full records with hierarchy information or import hierarchy information for existing records
P-11.3.	System will check that each item type can physically fit into the next item type in the hierarchy (e.g., height and weight of “eaches” do not exceed height and weight of “case”)
P-11.4.	User must be able to change the item level on a record while creating a hierarchy
P-11.5.	If the status of a record within a hierarchy changes, the user should be notified if it will affect the status of other records within the hierarchy
P-12.	Export and print hierarchies in a user-friendly format
P-13.	Generate and view X-Customer Standard supported barcodes of various types and sizes
P-14.	Export and print X-Customer Standard supported barcodes in standard image formats
P-15.	Create, save, export, and print Product Information Sheets for each record with all record attributes and images
P-15.1.	Users can select from and edit multiple page layout templates for product sheets
P-16.	Create a digital GIN that can be imbedded in a web site
P-17.	Permission records for publishing; records will be viewed within the Data Access module
P-18.	Transfer record ownership to another member
P-19.	Use an external system (e.g. QuickBooks, SAP) to create and manage data and import it into the application solution

2.3. LOCATION REQUIREMENTS

The Location Module is used by data owners to create and manage LN data related to their own locations, and to permission data to be published in the Data Access module. Users include manufacturers or distributors who need to track where their products are going; the entities receiving the shipments (e.g., hospitals in the health care industry or restaurants in the food service industry); and group purchasing organization or other agents managing data on their behalf. Many users will use the application throughout their workday. High level requirements enable the user to:

ID	Requirement Description
L-1.	Create LN records individually, via import, through a step-by-step wizard, or by cloning an existing record. Records will include required and non-required attributes that may vary by industry
L-1.1.	Incomplete records can be saved as Draft
L-2.	Assign a LN to a record automatically or manually at any point during record creation; a check-digit will be automatically assigned to the LN
L-2.1.	Some industries automatically assign LNs from a shared pool
L-2.2.	Prefix licensee can opt for auto-assign or manual
L-2.3.	Prefix licensee can reserve LNs to hold out from auto-assign
L-3.	Set a preference for how LNs will be assigned (e.g., auto-assign or manual)
L-4.	Edit record attributes before finalizing
L-5.	Edit record attributes, manually or via import
L-5.1.	If a LN changes but the location attributes do not, a reference should be made back to the old number
L-6.	View, filter, and sort all records in a single view
L-6.1.	Select individual records to take any action (e.g., edit, export, etc.)
L-6.2.	Select multiple records to take action and apply the same action to all selected records
L-7.	View, filter, and sort all LNs in a single view
L-8.	View potential duplicate records and easily remove or edit duplicate records
L-8.1.	System will alert user of possible duplicate records during creation process as well as run duplicate reports for completed records
L-9.	View and manage the status of a record or LN
L-9.1.	The system or user will assign a status to each LN; LN status may update to reflect record status
L-9.2.	The system or user can assign a status to each record
L-9.3.	If a LN is no longer in use, display the date taken out of use and when it will be available for re-use; rules may vary by industry
L-10.	Create, edit, and view a hierarchy of LNs manually, via import, or via a visual format such as drag-and-drop
L-10.1.	Hierarchies can have an unlimited number of levels and be organized by various attributes

L-10.2.	Imports can include full records with hierarchy information or import hierarchy information for existing records
L-10.3.	If the status of a record within a hierarchy changes, the user should be notified if it will affect the status of other records within the hierarchy
L-11.	Export and print hierarchies in a user-friendly format
L-12.	Permission records for publishing; records will be viewed within the Data Access module
L-13.	Record annual verification of records
L-14.	Transfer record ownership to another member
L-15.	Use an external system (e.g. QuickBooks, SAP) to create and manage data and import it into the application solution

2.4. ACCESS DATA REQUIREMENTS

The Access Data Module allows users to search for, view, and export Prefix, GIN, and LN data. It is the “subscribe” side of the publish-and-subscribe model detailed under General Requirements above. This module has a very broad user base that includes but is not limited to supply chain partners verifying large quantities of location data and large retailers accessing data through the API to verify prefix licensees. For some users, this will be their primary work tool. User base may also include non-members looking up information for a single business.

- Clients (end-users) subscription will determine which data can be viewed and which export options are available.
- Non-members (i.e., the general public) can search a limited data set, but can not export.

X-Customer-owned data as well as data that have been “published” by the data owner can be accessed. There is no editing functionality in this module. High level requirements enable the user to:

ID	Requirement Description
D-1.	Search for Prefix, GIN, and LN records
D-1.1.	Several fields will be available for search (e.g., Name, City)
D-1.2.	Advanced search and filter options will be available
D-2.	Request access to a basic or full record (i.e. subscribe to the record)
D-2.1.	Request will be sent to the data owner who must approve before the data becomes available
D-3.	Request to be added to a controlled group
D-4.	View basic and full records and their hierarchies
D-5.	Export one or more records to any format detailed in General Requirements above
D-6.	Print one or more records
D-7.	Perform all functions listed above via web interface or API
D-8.	Easily access information on how to pay for ad hoc access to data
D-8.1.	The payment process will be handled outside of the integrated application solution and is not in scope for this project

2.5. TECHNICAL REQUIREMENTS

ID	Requirement Description			
NFR-1.	X-Customer does not constrain the vendor in proposing technology platforms.			
NFR-2.	To support the need for increased data access, supplier should propose relevant technology to leverage data marts (non-transactional). The system should also be scalable in both the transaction and non-transaction databases to account for the projected growth. Data latency is currently overnight but should support near real-time updates. Current Database Record Counts and Projected Growth by 2024			
	Database	Current Records	Current Size	Future Records
	Company Prefixes	500 000	5 GB	700 000
	Product (GIN)	10 000 000	3 GB	70 000 000
	Location (LN)	550 000	2 GB	3 500 000
NFR-3.	To provide secure, streamlined access to the application solution, X-Customer requires that the new application solution support claims-based authentication and authorization. User accounts and roles will be defined in an Enterprise Identity Management system to enable Single Sign-On (SSO). The Identity Management solution has not been finalized but will support claims-based authentication standards such as OAuth and SAML.			
NFR-4.	The architecture should be developed using a service-oriented architecture. Web services using REST (preferable) or SOAP should be used to integrate functionality and data across the application suite instead of direct database access or copying data. REST-based APIs will also be available to members and should be compatible with an API management solution (currently 3scale).			
NFR-5.	The solution should adhere to technical standards as specified by X-Customer including (but not limited to) coding standards, database connection standards, database naming standards, integration standards and the standard for approved technologies and products. If the solution deviates from any X-Customer technical standards, an impact analysis should be conducted by project core team members at no cost to X-Customer. The impact analysis should include the technical, operational, support and financial assessments of the proposed change.			
NFR-6.	To enable future application enhancements, the architecture should be designed to be “cloud-ready.”			
NFR-7.	The application solution should be web-based. The solution should be built using industry standard and/or best practice web design patterns, frameworks, and components. X-Customer prefers solutions that are built on or highly leverage industry standard or X-Customer supported design patterns, frameworks, and components in order to improve maintainability and lower maintenance costs. No client-side software installation (e.g., Flash) should be required. The current applications support roughly 38,000 users and it is anticipated that by 2024, the number of users could grow to approximately 256,000 users, with up to 10% of those users accessing the application solution concurrently.			
NFR-8.	The solution should be highly available. The Service Level Agreement for all applications is 99.9%. Respondents should provide a plan to develop and test a failover process that will achieve high-availability. This includes the identification of hardware and software needs to support a failover. The failover process should have "failover monitoring" that will provide the ability to notify administrators of a failure.			

NFR-9.	To support business continuity, responses should include backup and disaster recovery procedures that comply with X-Customer policies and procedures.
NFR-10.	The solution should include appropriate security configuration, processes, and procedures. Responses should include the security design and any components, frameworks, libraries, and/or tools used to ensure the security of the application solution.
NFR-11.	The solution should include deployment to four environments: development, test, staging/pre-production, and production. Appropriate deployment processes should be included.
NFR-12.	Responses should include migrating the current application data, including users, to the new solution.
NFR-13.	The solution should be accessible on the following web browsers: • Internet Explorer 9 and above • Chrome current and previous version • Firefox current and previous version • Safari current and previous version • Edge current and previous version
NFR-14.	The solution should support the response times below assuming high-speed/broadband access: • Login to the application < 2 seconds 95% of the time @ 250 concurrent user load with no think time • General user navigation events < 1 second 95% of the time @ 250 concurrent user load with no think time • Transactions (saves, generate forms, etc.) < 2 seconds 95% of the time @ 50 concurrent user load with no think time • Searches < 3 seconds 95% of the time @ 150 concurrent user load with no think time • API response < 1 second 95% of the time @ 250 concurrent user load with no think time
NFR-15.	Exceptions may be made for some time-consuming transactions (e.g., when generating a barcode), however the performance of the system is extremely important to the user experience. Responses should include the appropriate performance testing processes to ensure the response times are met at the appropriate user load.
NFR-16.	The solution should support business process workflows including: • Record approval - Records that are entered into the system, either through the UI or through an import, can be submitted for approval, reviewed, and rejected or approved • Record validation - Records that contain a numeric identifier (GIN, LN), which can be auto-generated by the system or manually entered, should be validated to ensure compliance with X-Customer Standards • Record verification - Record attributes that are entered into the system, either through the UI or through an import, should be verified for accurate information
NFR-17.	The solution should provide near-real time event notifications when system errors occur.
NFR-18.	The solution should provide near-real time event notification when critical business functions fail.

NFR-19.	The solution should provide detailed logging for troubleshooting and process verification.
NFR-20.	The solution should generate an audit trail of all activity taking place at any point in the system.
NFR-21.	The solution (within reasonable accommodation) should be WCAG 2.0 Level A compliant to be accessible to users with disabilities.
NFR-22.	Responses should identify all third party solutions or open source libraries required for the solution and the license associated with each.

3. PROPOSAL GUIDELINES*

Proposal Submissions

Each Vendor assumes full responsibility for delivery and deposit of their complete proposals by the proposal due date. The X-Customer is not responsible for any loss or delay with respect to delivery of the proposal. Any proposal received after the date and time specified may not be considered. Once submitted, the proposals become the property of the X-Customer and cannot be returned. The X-Customer shall not be held liable for any activity or costs incurred by Vendors in the preparation of their proposals. Likewise, the X-Customer shall not be held liable for costs incurred during any negotiations on proposals or proposed contracts or for any work performed in connection therewith.

Proposal Life

The prices and other terms offered by Vendors must be firm for an acceptance period of 120 days from the final RFP response date.

Rejection of Proposals

The X-Customer reserves the right in its sole discretion to reject any or all proposals in whole or in part without incurring any cost or liability whatsoever. All proposals will be reviewed for completeness of the submission requirements. A proposal may be rejected if it fails to meet a material requirement of the RFP, if it is incomplete, or if it contains irregularities. A deviation is material to the extent that a proposal is not in substantial accord with RFP requirements.

Withdrawal of Proposals

Vendors may withdraw their proposals from consideration at any time prior to award of the contract. Vendors who decide to withdraw their proposal are asked to notify the X-Customer immediately of such withdrawal in writing directed to the point of contact in order to avoid unnecessary expenditure of resources in reviewing such proposals. The X-Customer shall have the right to destroy or retain for its records all copies of proposals withdrawn from consideration.

Request for Proposal is not a guarantee of contract

This Request for Proposal does not commit the X-Customer to awarding a contract. Vendor shall bear all costs incurred in the preparation of the proposal and participating in the proposal evaluation process. The X-Customer reserves the right to reject any and all proposals or to accept the proposal it considers most favorable to the X-Customer's interest in its sole discretion. The X-Customer further reserves the right to seek new proposals when such procedure is considered by it to be in the best interest of the X-Customer.

Conflicts

The Vendor must warrant and represent that there is nothing in the solution represented by the response to the RFP that presently conflicts or will conflict with any other duty or agreement of the Vendor. In particular, the Vendor must warrant that nothing in the RFP response violates any other agreement or duty of non-disclosure that the Vendor does not have the authority to reprint in the response or utilize in the implementation of the solution.

Relationships of Parties

Neither Party is an agent for the other and has no authority to represent or bind the other Party as to any matters, except as expressly authorized. Nothing contained in this RFP will be deemed or construed by the parties, nor by any third party, as creating any partnership or joint venture relationship between the parties. Nothing in this RFP will constitute or permit either party to execute on behalf of the other any contract or other document, or to bind or otherwise obligate the other party. The X-Customer has the sole right and obligation to supervise, manage, contract, subcontract, direct, procure, perform or cause to be performed, all work to be performed in response to the RFP.

4. STRUCTURE OF RFP RESPONSE

The Vendor must provide, as a separate Microsoft Word document, a file entitled “Vendor-Name Proposal.docx” (Vendor: please replace ‘Vendor-Name’ with your company’s name or acronym). Optionally, the Vendor may choose to separate each of the sections below into separate Word documents that follow the same naming convention.

The document must contain the following sections/topics:

Executive Summary

The Executive Summary should summarize the Vendor’s offering, its approach, and the value provided by the Vendor. It should be written to communicate to the X-Customer’s executive management team.

The Executive Summary must not exceed one (1) page in length (excluding title page), and no appendices or specific references to additional information will be accepted.

Solution Approach

The Solution Approach should cover the overall proposed solution and provide the main opportunity for the Vendor to impress on the completeness and capability of their solution. It should address functional and non-functional requirements and contain a high-level technical architecture and description of the proposed solution. In addition to the high level architecture, this section might describe other technical details including but not limited with the following:

- Solution overview and how key goals are achieved
- Architecture components
- Integration with external systems
- List of platforms, technologies and tools including type (name) of storage(s)
- Deployment approach
- Dependencies, Risks and Assumptions

Delivery Approach

The Delivery Solution should cover the following aspects:

- Used methodology
- Estimations
- Timeline, team structure
- Pricing including the total cost

5. EVALUATION PROCESS AND CRITERIA*

Following the initial review and screening of the written Proposals, using the selection criteria described below, several Vendors may be invited to participate in the final selection process, which may include participation in an oral interview, live demonstration, and/or submission of any additional information as requested by X-Customer.

During the evaluation process, X-Customer may require a Vendor's representative to answer questions with regard to the proposal and/or require certain Vendors to make a formal presentation to the evaluation team.

The following criteria will be used in evaluating the proposals:

1. Responsiveness of the proposal to the submission requirements set forth in the RFP
2. The feasibility and overall robustness of the solution including the user interface
3. The technical ability, capacity, and flexibility of the Vendor to perform the contract in a timely manner and on budget, as verified by, e.g., the quality of any demonstration, client references, demonstrated success in projects with similar requirements and any other contracts with X-Customer
4. The financial viability of the Vendor as evidenced by standard financial statements
5. The cost of the proposed solution